

CHARLEVOIX MRC, QC, Canada

WMO: 713190

Lat: 47.284N

Lon: 70.637W

Elev: 723

StdP: 92.94

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.3	(c) -25.7	(d) -32.8	(e) 0.2	(f) -27.9	(g) -30.0	(h) 0.3	(i) -25.3	(j) 8.4	(k) -12.6	(l) 7.4	(m) -12.0	(n) 2.1	(o) 310	(p) 0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	24.4	17.9	22.8	16.8	21.4	16.0	19.5	22.5	18.5	21.3	17.5	20.2	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	14.6	21.0	17.4	13.6	20.0	16.5	12.8	19.1	59.0	22.7	55.6	21.2	52.2	20.2	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.3	6.3	5.5	-31.8	27.4	2.3	1.7	-33.5	28.6	-34.8	29.6	-36.1	30.5	-37.7	31.7
			-32.0	21.5	2.3	1.2	-33.7	22.3	-35.0	23.0	-36.3	23.7	-38.0	24.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.1	-14.3	-12.9	-8.0	-0.6	6.9	12.2	15.1	14.3	10.4	3.2	-3.5	-10.3		
	DBStd	11.49	6.59	6.35	6.29	4.64	4.24	3.94	2.99	3.14	4.28	4.48	5.18	6.03		
	HDD10.0	3712	752	642	558	320	117	20	1	3	47	216	404	631		
	HDD18.3	6305	1010	876	816	569	355	187	108	129	241	469	654	889		
	CDD10.0	468	0	0	1	1	21	87	158	138	58	5	0	0		
	CDD18.3	19	0	0	0	0	1	4	7	6	2	0	0	0		
	CDH23.3	102	0	0	0	0	6	27	39	21	9	0	0	0		
	CDH26.7	6	0	0	0	0	0	2	3	1	1	0	0	0		
(14) Wind	WSAvg	2.6	2.5	2.6	2.9	2.9	2.6	2.5	2.4	2.3	2.4	2.7	2.6	2.6		
(15) Precipitation	PrecAvg	1215	90	75	81	96	105	109	123	112	108	109	104	100		
	PrecMax	1486	175	163	134	223	181	198	250	204	214	208	181	199		
	PrecMin	970	30	35	27	30	38	41	18	45	57	32	26	40		
	PrecStd	120	38	29	28	48	31	37	47	39	32	40	33	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	3.8	10.5	15.4	24.0	26.1	26.6	25.5	24.5	17.8	11.1	5.3		
		MCWB	4.9	1.8	6.4	9.2	15.7	18.5	19.3	18.8	18.5	12.2	8.2	5.0		
	2%	DB	1.2	0.9	5.4	11.7	20.5	23.7	24.2	23.5	21.6	14.7	8.0	1.8		
		MCWB	0.8	-0.3	2.5	6.3	12.3	16.8	18.4	17.8	17.0	11.4	5.9	1.3		
	5%	DB	-1.8	-1.3	2.7	9.0	17.9	21.6	22.6	21.7	19.6	12.4	5.9	-0.1		
		MCWB	-2.3	-2.4	0.3	4.7	11.4	15.4	17.1	16.6	16.1	9.7	4.3	-0.6		
	10%	DB	-4.7	-3.2	0.7	6.5	15.2	19.7	21.1	20.3	17.6	10.2	3.7	-2.0		
		MCWB	-5.3	-4.1	-1.0	2.8	10.2	14.5	16.3	16.0	14.9	7.9	2.4	-2.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	2.8	8.3	10.6	17.2	20.1	21.3	20.2	19.5	14.1	9.8	5.0		
		MCDB	4.6	3.1	10.1	13.7	21.4	24.3	24.7	23.4	22.7	15.9	10.2	5.3		
	2%	WB	0.5	-0.4	2.9	7.8	14.6	18.3	19.8	18.9	18.2	11.8	6.7	1.5		
		MCDB	0.8	0.5	4.7	10.2	18.4	21.4	22.5	21.8	20.8	13.9	7.4	1.7		
	5%	WB	-2.6	-2.3	0.7	5.1	12.7	16.9	18.7	18.0	16.8	10.1	4.4	-0.6		
		MCDB	-2.1	-1.5	2.2	8.5	15.7	19.9	21.3	20.6	18.7	12.0	5.2	-0.3		
	10%	WB	-5.4	-4.1	-0.8	3.1	10.7	15.5	17.5	17.0	15.2	8.2	2.4	-2.6		
		MCDB	-4.9	-3.3	0.8	6.0	14.4	18.6	20.2	19.3	17.2	9.7	3.4	-2.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	9.9	9.6	8.5	10.2	10.0	9.4	9.4	9.2	7.3	6.7	8.1		
		MCDBR	10.4	10.0	10.4	12.2	15.2	12.9	11.3	11.0	10.9	10.8	8.9	8.8		
	5% WB	MCWBR	10.3	9.0	8.2	8.2	8.7	7.4	6.5	6.6	7.1	7.9	7.5	8.9		
		MCWBR	10.3	9.3	9.3	11.0	12.1	10.3	9.5	9.4	9.6	9.8	8.0	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.237	0.254	0.285	0.317	0.325	0.348	0.364	0.346	0.328	0.303	0.276	0.253		
	taud		2.386	2.323	2.311	2.380	2.457	2.419	2.387	2.447	2.492	2.559	2.502	2.390		
	Ebn at Noon		877	927	942	939	938	914	893	898	886	863	820	805		
	Edh at Noon		67	89	107	111	107	112	114	103	90	72	61	60		
(44) All-Sky Solar Radiation	RadAvg		1.38	2.29	3.46	4.44	5.06	5.45	5.29	4.60	3.59	2.11	1.21	0.94		
	RadStd		0.18	0.28	0.28	0.37	0.53	0.36	0.55	0.30	0.29	0.18	0.14	0.09		

Historical Trends

		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (33 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page